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Activity - 5

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D Input {437}, 1323, 6173, 4199, 4344, 9679, 1989 and hash func, $h(x) = x \bmod 10$

- a) separate hash
- b) linear probing
- c) quadratic probing
- d) second hash

$$h_2(x) = 7 - (x \bmod 7)$$

Solution : separate hash

$$h(x) = x \bmod 10 \rightarrow \text{Hash table}$$

$$h(437) = 437 \bmod 10 = 7$$

$$h(1323) = 1323 \bmod 10 = 3$$

$$h(6173) = 6173 \bmod 10 = 3$$

$$h(4199) = 4199 \bmod 10 = 9$$

$$h(4344) = 4344 \bmod 10 = 4$$

$$h(9679) = 9679 \bmod 10 = 9$$

$$h(1989) = 1989 \bmod 10 = 9$$

Hash table:

0	
1	→ [437]
2	
3	→ [1323] → [6173]
4	→ [4344]
5	
6	
7	
8	
9	→ [4199] → [9679] → [1989]

linear probing:

$$h(\text{key}) = h(\text{key} + i)$$

4371, 1323, 6173, 4199,
4344, 9679, 1989.

soln:

$$\begin{aligned} \bullet h(4371) &= 2 \% 10 \\ &= 4371 \% 10 \\ &= \textcircled{1} \end{aligned}$$

$$\begin{aligned} \bullet h(1323) &= 1323 \% 10 \\ &= \textcircled{3} \end{aligned}$$

$$\begin{aligned} \bullet h(6173) &= 1323 \% 10 \\ &= 3 \text{ (collision)} \end{aligned}$$

$$h(\text{key}) = 6173 + 1 = \textcircled{4}$$

$$h_2(\text{key}) = \text{key}$$

$$\begin{aligned} \bullet h(4199) &= 4199 \% 10 \\ &= \textcircled{9} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= h(4199 + 1) \\ &= h(4200) \\ &= \textcircled{0} \end{aligned}$$

$$\begin{aligned} \bullet h(4344) &= 4344 \% 10 \\ &= 4 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= h(4344 + 1) \\ &= h(4345) \\ &= \textcircled{5} \end{aligned}$$

$$\begin{aligned} \bullet h(9679) &= 9679 \% 10 \\ &= 9 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= h(9679 + 1) \\ &= h(9680) \\ &= 9680 \% 10 \\ &= \textcircled{0} \text{ (collision)} \end{aligned}$$

$$h(\text{key}) = h(9679 + 2)$$

$$\begin{aligned} \bullet h(1989) &= 1989 \% 10 \\ &= 9 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} \bullet h(\text{key}) &= h(1989) + 1 \\ &= 1990 = 0 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} \bullet h(\text{key}) &= h(1989) + 2 \\ &= 1991 = 1 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} \bullet h(\text{key}) &= h(1989) + 3 \\ &= 1992 = \textcircled{2} \end{aligned}$$

Hash table:

0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	
7	
8	
9	4199

Quadratic probing

$$h(\text{key}) = h(\text{key} + i^2)$$

Soln:

$$\begin{aligned} \bullet h(4371) &= 4371 \% 10 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \bullet h(1323) &= 1323 \% 10 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \bullet h(6173) &= 6173 \% 10 \\ &= 3 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 6173 + 12 \\ &= 6174 \% 10 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \bullet h(4199) &= 4199 \% 10 \\ &= 9 \end{aligned}$$

$$\begin{aligned} \bullet h(4344) &= 4344 \% 10 \\ &= 4 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 4344 + 12 \\ &= 4345 \\ &= 5 \end{aligned}$$

$$\begin{aligned} \bullet h(9679) &= 9679 \% 10 \\ &= 9 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 9679 + 12 \\ &= 9680 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \bullet h(1989) &= 1989 \% 10 \\ &= 9 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 1989 + 12 \\ &= 1990 \text{ (collision)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 1989 + 22 \\ &= 1989 + 4 \\ &= 1993 \\ &= 3 \text{ (collision)} \end{aligned}$$

$$\begin{aligned} h(\text{key}) &= 1989 + 3^2 \\ &= 1989 + 9 \\ &= 1998 \\ &= 8 \end{aligned}$$

Hash table :

0	9679
1	4371
2	6023
3	823
4	6173
5	4344
6	
7	
8	1989
9	4199

//

Double hash:

$$h(\text{key}) = h_1(\text{key}) + (h_2(\text{key}) * i)$$

$$h_1 = x \bmod 10$$

$$h_2 = 7 - (x \bmod 7)$$

$$\bullet h(4371) = 4371 \% 10 = \textcircled{1}$$

$$\bullet h(1323) = 1323 \% 10 = \textcircled{3}$$

$$\bullet h(6173) = 6173 \% 10$$

$$h_1 = 3 \text{ (collision)}$$

$$h_2 = 7 - (6173 \% 7)$$

$$= 7 - 6$$

$$= 1 \text{ (collision)}$$

$$h(\text{key}) = 3 + (1 * 1) = \textcircled{4}$$

~~$$\bullet h(9679)$$~~

$$\bullet h(4199) = 4199 \% 10 = \textcircled{9}$$

$$\bullet h(4344) = 4344 \% 10$$

$$h_1 = 4 \text{ (collision)}$$

$$h_2 = 7 - (4344 \% 7)$$

$$= 7 - 4$$

$$= 3$$

$$h(\text{key}) = 4 + (3 * 1) = \textcircled{7} \text{ (collision)}$$

$$\bullet h(9679) = 9679 \% 10$$

$$h_1 = 9 \text{ (collision)}$$

$$h_2 = 7 - (9679 \% 7)$$

$$= 7 - 5$$

$$h(\text{key}) = 9 + (2 * 1) = 11 = 1 \text{ (collision)}$$

$$h(\text{key}) = 9 + (2 * 2) = 13 \text{ (collision)}$$

$$= 3$$

$$h(\text{key}) = 9 + (2 * 3) = 15 = \textcircled{5}$$

$$\bullet h(1989) = 1989 \% 10$$

$$h_1 = 9 \text{ (collision)}$$

$$h_2 = 7 - (1989 \% 7)$$

$$= 7 - 1 = 6$$

$$h(\text{key}) = 9 + (6 * 1)$$

$$= 15 \text{ (collision)}$$

$$= 5$$

$$h(\text{key}) = 9 + (6 * 2)$$

$$= 21 \text{ (collision)}$$

$$= 1$$

$$h(\text{key}) = 9 + (6 * 3)$$

$$= 9 + 18$$

$$= 27$$

$$= \textcircled{7} \text{ (collision)}$$

$$h(\text{key}) = 9 + (6 * 4)$$

$$= 33 \text{ (collision)}$$

$$h(\text{key}) = 9 + (6 * 5)$$

$$= 39 \text{ (collision)}$$

$$h(\text{key}) = 9 + (6 * 6)$$

$$= 45 = 5 \text{ (collision)}$$

$$h(\text{key}) = 9 + (6 * 7)$$

$$= 51 = 5 \text{ (collision)}$$

$$h(\text{key}) = 9 + (6 * 8)$$

$$= 57 \text{ (collision)}$$

• Key - 1989 : cannot be inserted
hash table:

0	
1	4371
2	
3	1323
4	6173
5	9679
6	
7	4344
8	
9	4199